

7-Day SQL Roadmap for Data Analyst Interviews

The goal is not to learn every SQL concept, but to master the topics that appear in most data analyst interviews.

Day 1 – SQL Basics

Learn: What is SQL, Databases vs Tables, Rows vs Columns, Primary Key, Foreign Key, Data Types.

Commands: SELECT, DISTINCT, LIMIT/TOP, ORDER BY, WHERE, AND, OR, NOT, IN, BETWEEN, LIKE, IS NULL.

Practice: Solve 20 easy SQL questions.

Goal: Become comfortable retrieving and filtering data.

Day 2 – Filtering & Data Transformation

Learn: Arithmetic Operators, Aliases (AS), CASE WHEN.

String Functions: CONCAT, UPPER, LOWER, SUBSTRING, LENGTH.

Date Functions: CURRENT_DATE, YEAR, MONTH, DAY, DATEDIFF.

Practice: Solve 15-20 SQL questions.

Goal: Learn to clean and transform data.

Day 3 – Aggregations

Learn: COUNT(), SUM(), AVG(), MAX(), MIN().

Concepts: GROUP BY, HAVING.

Practice Questions: Average salary by department, Revenue by month, Total customers, Highest sales, Category-wise revenue.

Practice: 20 questions.

Goal: Learn to summarize data.

Day 4 – SQL Joins (Most Important)

Learn: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, SELF JOIN (Basic).

Understand: One-to-One, One-to-Many, Many-to-Many relationships.

Practice: Orders with customer names, Employees with departments, Customers without orders, Products without sales. Solve 20 join questions.

Goal: Combine data from multiple tables confidently.

Day 5 – Intermediate SQL

Learn: Subqueries, Correlated Subqueries, CTEs, Views (Basics), UNION, UNION ALL.

Practice: Solve 15-20 questions.

Goal: Break complex problems into smaller, readable queries.

Day 6 – Window Functions

Learn: ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), LEAD(), OVER(), PARTITION BY, Running Total, Rolling Average.

Practice: Second highest salary, Top 3 products, Previous month's sales, Consecutive dates, Duplicate records. Solve 15-20 questions.

Goal: Master advanced SQL interview questions.

Day 7 – Mock Interview & Revision

Choose one dataset (Sales, E-commerce, HR, Netflix, etc.).

Solve: 5 Easy, 10 Medium, 5 Hard questions.

Revise: Joins, GROUP BY, Window Functions, CASE WHEN, CTEs, Subqueries.

Also: Explain every query aloud, review mistakes, optimize queries.

Goal: Think like a Data Analyst, not just someone who writes SQL.

Daily Routine (2-3 Hours)

30 min Learn concepts

60-90 min Solve SQL problems

20 min Review mistakes
15 min Revise previous day's topics

High-Priority Topics

- ★★★★ Window Functions
- ★★★★ INNER JOIN & LEFT JOIN
- ★★★★ GROUP BY & HAVING
- ★★★★ Aggregate Functions
- ★★★★ CASE WHEN
- ★★★★ CTEs
- ★★★★ Subqueries
- ★★ Date & String Functions
- ★★ UNION / UNION ALL
- ★★ SELECT, WHERE, ORDER BY

Best SQL Practice Platforms

SQLBolt, DataLemur, StrataScratch, LeetCode SQL, HackerRank SQL.

Target

Solve 100–150 well-chosen SQL questions, understand every solution, and explain your approach confidently. That's enough to prepare for most entry-level Data Analyst SQL interviews.